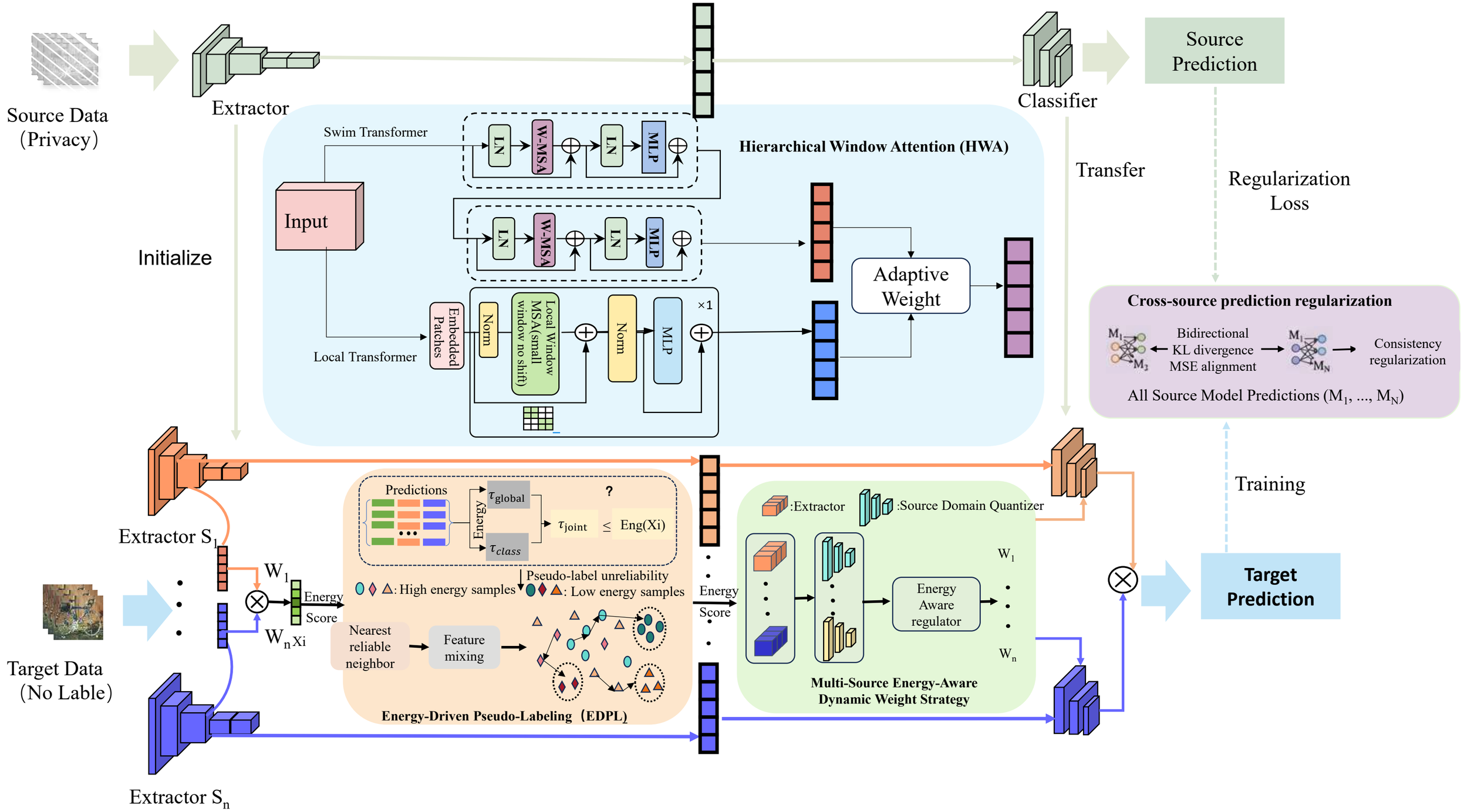
Source Data
(Privacy)
Extractor
Classifier
Source
Prediction
Transfer
Regularization
Loss
Initialize
Swim Transformer
LN
W-MSA
LN
MLP
Hierarchical Window Attention (HWA)
Input
Adaptive
Weight
Embedded
Patches
Norm
Local Window
MSA(small
window no shift)
Norm
MLP
×1
Local Transformer
Cross-source prediction regularization
Bidirectional
KL divergence
MSE alignment
Consistency
regularization
All Source Model Predictions (M_1, ..., M_N)
Training
Extractor S_1
Extractor S_n
Target Data
(No Lable)
W_1
W_n Xi
Energy
Score
Predictions
Energy
τ_{global}
τ_{class}
τ_{joint}
≤
Eng(Xi)
?
Pseudo-label unreliability
: High energy samples
: Low energy samples
Nearest
reliable
neighbor
Feature
mixing
Energy-Driven Pseudo-Labeling (EDPL)
Energy
Score
:Extractor
:Source Domain Quantizer
Energy
Aware
regulator
W_1
W_n
Multi-Source Energy-Aware
Dynamic Weight Strategy
Target
Prediction